

School of Computer Science Engineering and Technology

Lab No. - 11

Total Marks—1

Course-B. Tech.	Type- Core
Course Code- CSET301	Course Name- Artificial Intelligence and Machine Learning
Year- 2025	Semester- Odd
Date-	Batch- 2023-2027

CO-Mapping

	CO1	CO2	CO3	CO4	CO5	CO6
Q1		√		√		

Objective:

To implement **AdaBoost** and **Gradient Boosting Classifiers** (using Scikit-learn), perform binary classification on the Raisin Dataset, apply **hyperparameter tuning**, and compare their performance with each other.

Dataset:

Use the **Raisin Dataset** from:

<https://archive.ics.uci.edu/dataset/850/raisin>

Questions / Steps:

1. Data Pre-processing

- Load the dataset using Pandas and display the first 5 rows.
- Check for null/missing values. If present, handle them with a suitable approach.
- Convert the **Class** label into binary numeric values: Kecimen → 0, Besni → 1.
- Perform feature importance check using **Chi-Square test** and drop the least important feature(s).
- Split the dataset into **80% training and 20% testing** using `train_test_split`.

2. Implement AdaBoost Classifier

- a. Train an **AdaBoostClassifier** with default parameters.
- b. Evaluate the model on the test set using:
 - Confusion Matrix
 - Accuracy
 - Precision
 - Recall
- c. Perform **hyperparameter tuning** using GridSearchCV or RandomizedSearchCV.
 - Example hyperparameters:
 - `n_estimators` (50, 100, 200)
 - `learning_rate` (0.01, 0.1, 1)
 - `base_estimator` (Decision Stump vs deeper tree)
- d. Report the **best parameters** and the performance metrics (Accuracy, Precision, Recall, F1).

3. Implement Gradient Boosting Classifier

- a. Train a **GradientBoostingClassifier** with default parameters.
- b. Evaluate the model using the same metrics as above.
- c. Perform **hyperparameter tuning**:
 - Example hyperparameters:
 - `n_estimators` (50, 100, 200)
 - `learning_rate` (0.01, 0.1, 0.5, 1)
 - `max_depth` (3, 4, 5)
 - `subsample` (0.8, 1.0)
- d. Report the **best parameters** and performance metrics.

4. Comparative Analysis

- a. Compare **Decision Tree, AdaBoost, and Gradient Boosting** classifiers based on:
 - Accuracy
 - Precision
 - Recall
 - F1-score

- Training Time (optional)
 - b. Draw a **bar chart** to show the comparison of metrics.
 - c. Which model performs best for this dataset? Give a reasoning based on results.

Deliverables:

1. Python code for all steps (with comments).
2. Tables for **evaluation metrics** of each classifier.
3. A **comparison chart** (bar graph) of all classifiers.
4. A short conclusion: *Which algorithm performs better and why?*